Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Part 1: Course Information

Course: ISE 331: Computer Security Fundamentals

Semester: Spring 2026

Main Instructor name: Amir Rahmati

Co-Instructor name: Xinrui Yu

Course hours:

Class 01: Tuesday 2:00 PM - 3:35 PM, Thursday 8:00 AM - 9:35 AM, Week 1-14

Class 02: Tuesday 3:50 PM - 5:25 PM, Thursday 9:50 AM - 11:25 AM, Week 1-14

Office hours: Lixi Building 201-4

Mon	09:30 – 11:00, 11:30 – 12:30	Tue	11:30 – 12:30
Thu	11:30 – 12:30	Fri	09:30 – 11:00

Prerequisites: Basic understanding of Computer Architecture, Networks, and OS

Course Description:

Introduces the basic concepts and terminology of computer security. Covers basic security topics such as cryptography, operating systems security, network security, and language-based security.

Major Topics Covered:

- Basic Crypto
- System Security
- Network Security
- Web Security
- OS Security
- Mobile Security
- Forensics
- Human Factors
- Emerging Threats
- Authentication
- Access Control
- Privacy
- Cloud Security

Required Course Textbook and Materials:

Principles of Computer Security: CompTIA Security + and Beyond (5e, William Arthur Conklin, Greg White, Chuck Cothren, Roger Davis, Dwayne Williams, McGraw-Hill, 2018. ISBN: 978-1260026016)

Course Delivery: Lixi Building 403

How We Will Communicate:

Email: amir.rahmati@stonybrook.edu, xryu47@gmail.com

Part 2: Course Learning Objectives

Upon completion of the course, students will be able to:

1) Conversant with the basic terminology and concepts of computer security.
2) Understand basic security threats to systems and networks.
3) Be familiar with basic strategies used to protect systems and networks.
4) Be able to analyze, design, and build secure systems of basic complexity.

Part 3: Course Schedule

The Course Schedule is subject to change.

Week	Date	Quiz/Exam	Topic	Homework
1	Mar 03		Class Overview, Basic Crypto	-
	Mar 05		Symmetric Encryption	
2	Mar 10		Macs & Hashes	
	Mar 12	Quiz 1	Asymmetric Encryption	MD5 Collision Attack Lab
3	Mar 17		Authentication, PKI	
	Mar 19		SSL/TLS	
4	Mar 24	Quiz 2	Access Control	
	Mar 26		Web Browser, Web App	Cross-Site Request Forgery Attack Lab
5	Mar 31		Web App, Web Tracking	
	Apr 02	Quiz 3	DNS	
6	Apr 07		BGP	
	Apr 09		Worms and Other Malware	Morris Worm Attack Lab
7	Apr 14		Firewalls, IDS	
	Apr 16		Sports Event No Classes	
8	Apr 21	Midterm Exam	Shellcode	Firewall Evasion Lab
	Apr 23		Sandbox, Reverse Engineering	
9	Apr 28		Memory Errors	
	Apr 30	Quiz 4	ROP, Heap	Shellcode Development Lab
10	May 05		Kernel	
	May 07		Racing Conditions	
11	May 12	Quiz 5	Program Analysis	
	May 14		Symbolic Execution	Buffer-Overflow Attack Lab (Server Version)
12	May 19		Side Channels	
	May 21		Ethics	
13	May 26	Quiz 6	HW Security	Race Condition Vulnerability Lab
	May 28		ML Security	
14	Jun 02		IoT Security	

Part 4: Grading, Attendance, and Late Work Policies

In this course, you will be assessed on the following:

Composition of Grade	Percentage	Dates
Homework Assignments	30	● See course schedule above
Quizzes	20	● See course schedule above
Exam 1	20	● TBD
Exam 2	30	● TBD

- Attendance and Late Policy: Absence or lateness will affect your grade.
- Assignments and Tests: Due dates will be announced in class.
- Late Work Policy: ALL WORK MUST BE SUBMITTED N OR BEFORE THE DUE DATE AND TIME.
  - Assignments submitted late will lose 5 points per day.
  - Incomplete assignments or assignments that do not fulfill requirements will be returned and not counted.

Part 5: Academic Integrity Statement

- Each student must pursue his or her academic goals honestly and be personally accountable for all submitted work. Representing another person's work as your own is always wrong. Any form of academic dishonesty, including cheating and plagiarism, will be reported to the Institute.